



Acute kidney injury risk associated with piperacillin/tazobactam compared with cefepime during vancomycin therapy in hospitalised patients: a cohort study stratified by baseline kidney function

Nakyung Jeon^a, Ben Staley^b, Kenneth P. Klinker^c, Juan Hincapie Castillo^a, Almut G. Winterstein^{a,d,*}

^a Department of Pharmaceutical Outcomes and Policy, University of Florida, Gainesville, FL, USA

^b Department of Pharmacy, UF Health Shands Hospital, Gainesville, FL, USA

^c Department Pharmacotherapy and Translational Research, University of Florida, Gainesville, FL, USA

^d Department of Epidemiology, University of Florida, Gainesville, FL, USA

ARTICLE INFO

Article history:

Received 5 January 2017

Accepted 22 February 2017

Keywords:

Vancomycin

Piperacillin/tazobactam

Acute kidney injury

Renal impairment

Inverse probability of treatment weighting

IPTW

ABSTRACT

Recent studies have found an association between piperacillin/tazobactam when added to vancomycin and acute kidney injury (AKI) risk. However, studies were limited by the small sample size and residual confounding. The aim of this study was to compare the risk of AKI with vancomycin plus piperacillin/tazobactam (VPT) versus vancomycin plus cefepime (VC) and to examine whether pre-existing renal impairment mediates the risk. This was a retrospective cohort study using electronic health records for patients admitted to two hospitals in 2012–2013. The outcome, AKI, was defined as an increase in serum creatinine level of ≥ 0.3 mg/dL or $\geq 50\%$ from baseline. Patients were stratified by level of renal impairment as estimated by baseline creatinine clearance. Inverse probability of treatment weighting was used to balance baseline covariates between groups. Cox proportional hazards regression was used to evaluate VPT risk of AKI compared with VC. A total of 935 (17.53%) AKI cases were identified among 5335 patients receiving VPT or VC. VPT was associated with a higher risk of AKI relative to VC, with an adjusted hazard ratio (aHR) of 1.25 [95% confidence interval (CI) 1.11–1.42] in the total population and 1.70 (95% CI 1.44–2.02) in patients with normal baseline renal function. However, no elevated risk was found in patients with prior renal impairment (aHR = 0.81, 95% CI 0.65–1.01). VPT was associated with a higher risk of AKI relative to VC. The association was true in patients with normal renal function but not in those with pre-existing renal impairment.

© 2017 Elsevier B.V. and International Society of Chemotherapy. All rights reserved.

1. Introduction

Vancomycin, a glycopeptide antibiotic, remains the drug of choice for the treatment of severe Gram-positive infections, including methicillin-resistant *Staphylococcus aureus*, methicillin-resistant coagulase-negative staphylococci and non-vancomycin-resistant enterococci [1]. Vancomycin is primarily excreted through the kidneys with 90% by glomerular filtration and active tubular secretion [2]. The incidence of nephrotoxic events among patients receiving vancomycin has gradually increased, with reported rates of 5–43% depending on the study population or definition of acute kidney injury (AKI) [3]. The postulated mechanism of vancomycin-associated nephrotoxicity includes drug-induced oxidative stress and mito-

chondrial dysfunction or cell necrosis in the cells of the proximal renal tubule [3].

Piperacillin/tazobactam (TZP), a β -lactam/ β -lactamase inhibitor combination, is commonly used concomitantly with vancomycin as empirical therapy for several infections, including nosocomial pneumonia, abdominal infection, skin and soft-tissue infection (SSTI), sepsis and osteomyelitis. Although nephrotoxicity rates are low when used as monotherapy [4], higher incidence rates ranging from 16% to 37% have been observed when TZP is concomitantly used with vancomycin, suggesting the potential for an additive renal effect between the two agents [5–8].

However, the evidence is controversial, with three studies [9–11] not finding an AKI risk when comparing vancomycin plus piperacillin/tazobactam (VPT) therapy with vancomycin plus cefepime (VC) therapy or vancomycin monotherapy. Furthermore, all of the previous epidemiological studies were conducted in relatively small study populations (i.e. <300 patients) and some did not adequately adjust for baseline AKI risk, which may have confounded the results [5–7,9,10,12]. VPT-induced AKI has been defined

* Corresponding author. Department of Pharmaceutical Outcomes and Policy, University of Florida, Gainesville, FL, USA. Fax: +1 352 273 6270.
E-mail address: almut@cop.ufl.edu (A.G. Winterstein).

in past observational studies based on changes in serum creatinine (SCr) values using baseline SCr estimates determined on hospital admission [5,6,10]. However, baseline SCr measurements taken immediately before VPT initiation would be better suited to ascertain AKI events because they control for other renal insults that occurred between admission and treatment initiation, including the infectious process itself.

Finally, pre-existing renal impairment has been linked to a variety of adverse health outcomes, including a higher risk of developing AKI [13–17], yet to our knowledge there has been no study examining VPT safety specifically in patients with pre-existing renal impairment. Also, some studies [6,7] specifically excluded patients with pre-existing renal impairment. A larger cohort study including a stratified cohort of patients with renal impairment and employing a dynamic assessment of renal function to better isolate the impact of therapy is needed.

Based on previous findings on AKI risk among patients with chronic renal disease, we hypothesised that patients with pre-existing renal impairment will show a higher incidence of AKI in general, and a greater risk for VPT and VC combinations than patients with normal renal function [12,18,19]. To examine this hypothesis, a large electronic health records (EHRs) database from two University of Florida (UF)-affiliated hospitals was analysed, ensuring adequate sample size and availability of a broad range of AKI risk factors to balance baseline risk between the VPT group and the VC group.

2. Materials and methods

This comparative safety study was conducted in a retrospective cohort of patients admitted to and discharged from UF Health Shands from 1 January 2012 to 31 December 2013, and from UF Health Jacksonville from 1 March 2013 to 31 December 2013, using EHR data. The study was approved by the institutional review and privacy boards of the University of Florida (Gainesville, FL).

2.1. Inclusion and exclusion criteria

The study population consisted of patients aged ≥ 18 years who received at least three consecutive days of vancomycin during admission. Patients entered the study at the initiation of TZP or cefepime (index date) during vancomycin therapy. For inclusion in the study cohort, patients were further required to have at least one SCr value within 2 days prior to the index date and at least one follow-up SCr value after the index date in order to ascertain the onset of AKI. Patients were excluded if they: received both TZP and cefepime during follow-up; experienced severe blood loss defined as receipt of >6 units of packed red blood cells any time prior to the index date; had AKI on admission identified with diagnosis code ICD-9-CM: 584.x; had a baseline SCr value >4.0 mg/dL (stage 3 AKI); or had end-stage renal disease identified with diagnosis code ICD-9-CM: 585.6.

2.2. Outcomes measurement

The primary outcome, stage 1 AKI or higher, was an absolute increase in SCr by ≥ 0.3 mg/dL or $\geq 50\%$ from baseline during follow-up, according to KDIGO (Kidney Disease: Improving Global Outcomes) guidelines [20]. The secondary outcome was stage 2 AKI or higher defined as an increase in SCr of $\geq 100\%$. The SCr value closest to the index date was used to measure baseline renal function, and any follow-up value up to 3 days after drug discontinuation was considered to assess AKI.

2.3. Statistical analyses

2.3.1. Cox proportional hazards regression

Cox proportional hazards regression was used to estimate AKI risk of VPT compared with VC. Follow-up started the day after the index date. Patients were censored if VPT or VC had been discontinued for >3 days or the patient died or was discharged from the hospital, whichever came first.

2.3.2. Propensity score estimation and inverse probability of treatment weighting (IPTW) adjustment

Propensity scores that predict VPT versus VC exposure were estimated to summarise baseline confounding variables into a single score. Variables included: advanced age (>65 years); baseline SCr; hypoalbuminaemia (serum albumin <3.0 mg/dL); anaemia (haemoglobin <10 mg/dL), dehydration (blood urea nitrogen to SCr ratio >20); fever (body temperature >38.5 °C); hypotension (systolic blood pressure <80 mmHg or mean arterial blood pressure <60 mmHg); sex; race; cardiac surgery during index admission but preceding the index date; use of contrast agents; co-morbidities including heart failure, hypertension, liver disease, diabetes mellitus, chronic kidney disease (CKD) and myocardial infarction; principal diagnosis including pneumonia, sepsis, SSTI, osteomyelitis, urinary tract infection and bacteraemia; high-dose vancomycin (>4 g/day); use of angiotensin-converting enzyme inhibitors or angiotensin II receptor blockers; use of diuretics; use of non-steroidal anti-inflammatory drugs; and use of other high nephrotoxic medications (i.e. amikacin, amphotericin, cidofovir, cisplatin, colistin, cyclosporine, foscarnet, gentamicin, methotrexate, mitomycin, streptozocin, tacrolimus, tenofovir and tobramycin).

Baseline characteristics of the study populations were compared between groups using standardised differences, which are less sensitive to sample size than conventional *P*-values [21]. Using the propensity scores, the IPTW was calculated, which can balance measured baseline characteristics between groups to an extent similar to that achieved through random treatment allocation [22]. To reduce variance and to improve the precision of estimated exposure risk from IPTW analysis, stabilised weights were used [23].

2.3.3. Stratification

The pooled cohort was stratified by renal function to examine whether pre-existing renal impairment modifies AKI risk associated with VPT compared with VC. Patients with renal impairment were identified as those with a baseline creatinine clearance (CL_{Cr}) of ≤ 60 mL/min (estimated via the Cockcroft–Gault formula) or existence of CKD (ICD-9-CM: 585.x) upon hospital admission. Normal (non-impaired) renal function was defined as $CL_{Cr} > 60$ mL/min. Propensity scores and IPTW were estimated separately for each stratum, and AKI risk was assessed using separate Cox regression analyses. All analyses were conducted with SAS v.9.4 (SAS Institute, Cary, NC).

3. Results

Among 5335 patients included in the study cohort, 1980 (37.11%) were in the VPT group and 3355 (62.89%) patients were in the VC group. A total of 4530 patients (84.91%) received TZP or cefepime on the same day of vancomycin initiation. The median duration of vancomycin therapy was 5 days in both groups. The interquartile range was 3–7 days in the VPT group and 3–8 days in the VC group. Table 1 shows the baseline characteristics by exposure before and after IPTW adjustment. Before IPTW, more patients in the VPT group had hypoalbuminaemia, sepsis, liver disease, diabetes mellitus, SSTI, osteomyelitis and higher baseline SCr. In contrast, the control group was more likely to be located in the intensive care unit, to have experienced dehydration and hypotension, to have undergone cardiac surgery or to have had heart failure during the baseline period. After

Table 1

Baseline characteristics by exposure to vancomycin + cefepime (VC) versus vancomycin + piperacillin/tazobactam (VPT) before and after inverse probability of treatment weighting (IPTW) adjustment.^a

Characteristic	Unweighted			IPTW		
	VC (N = 3355)	VPT (N = 1980)	STDDIFF > 0.1	VC (N = 2677.0)	VPT (N = 2658.0)	STDDIFF > 0.1
Age >65 years	1119 (33.35)	549 (27.73)	*	841.6 (31.44)	829.9 (31.22)	
Race						
White	2426 (72.31)	1317 (66.52)	*	1877.3 (70.13)	1883.8 (70.88)	
Black	674 (20.09)	548 (27.68)		612.8 (22.89)	603.0 (22.68)	
Other	255 (7.60)	115 (5.81)		186.9 (6.98)	171.1 (6.44)	
Female sex	1414 (42.15)	829 (41.87)		1556.6 (58.15)	1563.4 (58.82)	
Intensive care	629 (18.75)	279 (14.09)	*	457.2 (17.08)	460.3 (17.32)	
Hypoalbuminaemia	959 (28.58)	640 (32.32)		812.7 (30.36)	828.2 (31.16)	
Dehydration	1793 (53.44)	812 (41.01)	*	1301.3 (48.61)	1297.4 (48.80)	
Hypotension	906 (27.00)	501 (25.30)	*	719.9 (26.89)	728.5 (27.41)	
Principal diagnosis						
Pneumonia	749 (22.32)	389 (19.65)		576.1 (21.52)	568.5 (21.39)	
Sepsis	688 (20.51)	512 (25.86)		610.6 (22.81)	610.8 (22.98)	
SSTI	349 (10.40)	492 (24.85)	*	425.3 (15.89)	417.1 (15.69)	
Osteomyelitis	189 (5.63)	223 (11.26)	*	208.7 (7.80)	206.6 (7.77)	
UTI	426 (12.70)	245 (12.37)		343.5 (12.83)	346.1 (13.02)	
Bacteraemia	125 (3.73)	65 (3.28)		94.5 (3.53)	91.1 (3.43)	
Cardiac surgery	33 (0.98)	3 (0.15)	*	17.9 (0.67)	11.5 (0.43)	
Contrast agent use	40 (1.19)	16 (0.81)		29.0 (1.08)	45.5 (1.71)	
Use of ACEIs/ARBs	493 (14.69)	361 (18.23)		424.6 (15.86)	416.1 (15.66)	
Use of diuretics	921 (27.45)	448 (22.63)	*	686.3 (25.64)	673.1 (25.32)	
Use of NSAIDs	1104 (32.91)	679 (34.29)		895.3 (33.44)	870.1 (32.74)	
Use of other NTXs ^b	268 (7.99)	176 (8.89)		227.3 (8.49)	218.9 (8.24)	
Anaemia	878 (26.17)	526 (26.57)		718.3 (26.83)	719.2 (27.06)	
Co-morbidities						
Heart failure	669 (19.94)	295 (14.90)	*	484.1 (18.08)	493.5 (18.57)	
Hypertension	1538 (45.84)	908 (45.86)		1224.6 (45.75)	1199.2 (45.12)	
Liver disease	168 (5.01)	146 (7.37)		164.8 (6.16)	161.6 (6.08)	
Diabetes mellitus	231 (6.89)	328 (16.57)	*	282.3 (10.55)	277.0 (10.42)	
Chronic kidney disease	406 (12.10)	242 (12.22)		334.5 (12.50)	340.9 (12.82)	
Myocardial infarction	75 (2.24)	44 (2.22)		64.4 (2.41)	60.7 (2.28)	
Baseline SCr (mean ± S.D.)	0.93 ± 0.54	1.02 ± 0.54	*	0.98 ± 0.54	0.98 ± 0.59	
High-dose vancomycin >4 g/day	237 (7.06)	113 (5.71)		171.0 (6.38)	149.2 (5.62)	
Vancomycin trough >20 µg/dL	64 (1.91)	51 (2.58)		61.7 (2.31)	71.0 (2.67)	

STDDIFF, standardised difference; SSTI, skin and soft-tissue infection; UTI, urinary tract infection; ACEI, angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; NSAID, non-steroidal anti-inflammatory drug; NTX, nephrotoxic medication; SCr, serum creatinine; S.D., standard deviation.

^a Data are *n* (%) unless otherwise stated.

^b Including amikacin, amphotericin, cidofovir, cisplatin, colistin, cyclosporine, foscarnet, gentamicin, methotrexate, mitomycin, streptozocin, tacrolimus, tenofovir and tobramycin.

* Indicates STDDIFF > 0.1.

IPTW, the VPT and VC groups had similar distributions of baseline covariates demonstrated by no baseline covariate having a standardised difference >0.1 (Table 1). Baseline characteristics of stratified cohorts by renal function are available in [Supplementary Tables S1 and S2](#).

A total of 935 (17.53%) patients developed stage 1 or higher AKI. A slightly larger proportion of patients in the VPT group developed AKI compared with the VC group [19.65% (389/1980) and 16.27% (546/3355), respectively]. Compared with VC exposure, the adjusted hazard ratio (aHR) associated with use of VPT was 1.25 [95%

confidence interval (CI) 1.11–1.42]. The AKI risk of VPT in 3807 patients with normal renal function was more pronounced, with an aHR of 1.70 (95% CI 1.44–2.02), whilst the risk was not significant in 1528 patients with pre-existing renal impairment (aHR = 0.81, 95% CI 0.65–1.01) (Table 2).

In the secondary analysis, similar results were found for 5.40% of patients (288/5335) who developed stage 2 AKI or higher, including 6.72% (133/1980) VPT patients and 4.62% (155/3355) VC patients. The aHR for VPT was 1.64 (95% CI 1.30–2.06). In the stratified analysis, the aHR was 0.94 (95% CI 0.55–1.52) in patients with $CL_{Cr} \leq 60$ mL/min and 2.17 (95% CI 1.65–2.87) in patients with $CL_{Cr} > 60$ mL/min, indicating existing renal impairment modified the VPT effect on AKI risk (Table 3).

Table 2

Crude and adjusted risk for stage 1 or higher acute kidney injury stratified by estimated baseline creatinine clearance (CL_{Cr}).

	Person-days	# outcome	cHR	aHR (95% CI)
All				
VC	30 593	546	Ref.	Ref.
VPT	17 182	389	1.28	1.25 (1.11–1.42) ^a
Impaired renal function (baseline $CL_{Cr} \leq 60$ mL/min)				
VC	7 563	259	Ref.	Ref.
VPT	5 486	143	0.76	0.81 (0.65–1.01)
Normal renal function (baseline $CL_{Cr} > 60$ mL/min)				
VC	23 030	287	Ref.	Ref.
VPT	11 696	246	1.70	1.70 (1.44–2.02) ^a

cHR, crude hazard ratio; aHR, adjusted hazard ratio; CI, confidence interval; VC, vancomycin + cefepime; VPT, vancomycin plus piperacillin/tazobactam.

^a $P < 0.05$.

4. Discussion

Using a large population of hospitalised patients and a broad spectrum of AKI risk factors to balance comparison groups, this study found that patients administered VPT have a greater risk of AKI compared with VC. Using consensus definitions for AKI, the increased nephrotoxicity of VPT was found to be present for both stage 1 and stage 2 AKI. It was further found that, in this study population, this AKI risk was driven by the VPT effect in patients with normal but not impaired baseline renal function.

These results have important clinical implications. The higher risk for stage 1 and 2 AKI in the combination therapy group stresses

Table 3

Crude and adjusted risk for stage 2 or higher acute kidney injury stratified by estimated creatinine clearance (CL_{Cr}).

	Person-days	# outcome	cHR	aHR (95% CI)
All				
VC	33 581	155	Ref.	Ref.
VPT	19 138	133	1.51	1.64 (1.30–2.06) ^a
Impaired renal function (baseline CL _{Cr} ≤ 60 mL/min)				
VC	8 434	56	Ref.	Ref.
VPT	6 058	30	0.75	0.94 (0.55–1.52)
Normal renal function (baseline CL _{Cr} > 60 mL/min)				
VC	25 147	99	Ref.	Ref.
VPT	13 080	103	2.00	2.17 (1.65–2.87) ^a

cHR, crude hazard ratio; aHR, adjusted hazard ratio; CI, confidence interval; VC, vancomycin + cefepime; VPT, vancomycin plus piperacillin/tazobactam.

^a $P < 0.05$.

the need for treatment individualisation in the selection of an appropriate antibiotic regimen. Antimicrobial stewardship not only emphasises the appropriate use of antibiotics for the prevention of microbial resistance and spread of infections, but also considers improvement of patient outcomes. The latter includes prevention of adverse drug events such as AKI, which can have debilitating effects on functional recovery of patients. In particular, stage 2 AKI had been associated with de novo CKD risk, prolonged hospital stay and increased mortality [24,25]. The direct association between stage 1 AKI represented by a small increase in SCr and its risk for adverse outcomes is less clear, in part due to direct harmful effects of its risk factors such as volume overload, electrolyte imbalance, metabolic acidosis and uraemia, which may increase mortality risk, hospital stay and costs [26]. Given the availability of alternative antibiotics and significant harm that could result from AKI, one must weigh the benefit of antimicrobial coverage against the potential for kidney damage.

To our knowledge, three studies have specifically evaluated the relationship between VPT and AKI against VC as an active comparator. The first, a propensity score-matched cohort study in 110 patients with normal baseline renal function found a significant VPT-associated AKI risk with an adjusted odds ratio of 5.67 (95% CI 1.66–19.33) [6]. The other two studies, both with samples of <200 patients, did not demonstrate an association but both lacked power to detect less than a doubling in AKI incidence as statistically significant [the HR for one study was 3.45 (95% CI 0.96–12.40)] [9,10].

The current study has sufficient sample size to examine the effect on more severe but less common AKI stages. With greater clinical significance of moderate to severe AKI (stage 2 or higher) than mild AKI (stage 1 AKI), the current findings highlight the importance of AKI prevention strategies such as avoiding nephrotoxic drugs and monitoring SCr and urine output in patients receiving VPT, or substituting VPT with VC [27,28].

This study did not confirm the initial hypothesis that patients with pre-existing renal impairment are more vulnerable to VPT nephrotoxic effects. In fact, no excess risk was found for either AKI stage 1 or 2 in VPT-exposed patients. Several alternative hypotheses may explain these findings but none could be addressed by the current analysis. First, patients with impaired renal function already have a higher risk for AKI and the contribution of VPT may be proportionally less pronounced than in patients with normal renal function. Second, the identification of AKI using an elevation in SCr may not be an adequately sensitive measurement in patients with underlying renal impairment. A recent study demonstrated that use of SCr increase to identify AKI is compromised by high false-positive rates, especially in patients with higher baseline SCr values, potentially misclassifying acute-on-chronic kidney injury in patients with CKD [29]. False-positives in both exposure groups would bias results to the null hypothesis (no difference) and thus may explain our in-

ability to discern VPT-associated AKI risk in patients with pre-existing renal impairment. Third, CKD patients may be more closely monitored to prevent AKI, and reduced exposure to other renal insults during therapy or additional preventive measures may have reduced the risk for AKI. In fact, pre-existing CKD was associated with better survival in hospitalised patients who developed AKI in previous studies [30]. Finally, other potentially nephrotoxic agents exhibit a similar counter-intuitive effect, including renin-angiotensin system antagonists, which are nephron-protective in the setting of CKD [31] whilst their use is associated with AKI in other clinical scenarios such as post-cardiac surgery or volume depletion [32]. Two randomised studies also found no significant difference in risk of renal toxicity among CKD patients between two types of contrast agents of which one is known to be associated with greater AKI risk than the other in patients without CKD [33,34].

Future studies evaluating drug-associated AKI should consider similar stratification by baseline renal function to allow further study of such effect-modifying properties. In addition, optimal strategies for managing VPT-associated AKI should be determined separately for patients with and without renal impairment. For example, whilst general efforts to minimise absolute AKI incidences may be important for both CKD patients, the choice of their vancomycin combination regimen may not be as critical as for those with normal function, where avoiding a VPT regimen would provide great opportunity to prevent AKI. Of note, hospital-acquired AKI has been found to increase the length of stay by an average of 4 days and to increase the cost of hospitalisation by ca. US\$7500 [35]. Clinicians must be cognizant of the risk factors that put individuals at higher risk for development of drug-induced AKI and monitor patients during the course of pharmacotherapy.

This study has several major strengths. We successfully considered a broad range of risk factors obtained from institutional EHRs, achieving excellent balance between comparison groups in terms of baseline renal risk. The large sample size of >5000 patients represents a diverse population from two different institutions that allowed an estimation of precise AKI risk estimates. Importantly, the sample size allowed examination of stage 2 AKI risk, a severity level of nephrotoxicity that is more relevant for risk–benefit considerations in selecting treatment for serious infections. It also allowed stratification by baseline renal function and generated an important hypothesis for future research that can further inform clinical decision-making in selecting the most efficacious and safe antibiotic regimen for patients with severe infection.

Study limitations should be noted in the interpretation of the presented results. First, owing to the observational design and despite a broad approach to capture confounders, unmeasured and residual confounding remains possible. It should be noted that both VPT and VC are standard empirical treatments and should be interchangeable. This was reflected by a similar distribution of baseline characteristics among comparison groups even before IPTW. We did not assess renal risk factors that may have emerged after drug initiation, representing an intention-to-treat approach. Future studies may use advanced analytical methods such as marginal structural models in order to control for time-varying covariates [36]. Finally, whilst we explored the effect-modifying effects of baseline renal function, other patient characteristics may also influence VPT risk and may not have been well represented in this two-hospital sample. Multicentre studies are needed to assess the consistency of the study findings in other inpatient cohorts.

5. Conclusion

This study demonstrated an association between TZP and AKI risk compared with cefepime among vancomycin-treated patients with normal renal function. VPT-associated AKI risk was not present among patients with pre-existing renal impairment. Multicentre

studies are needed to assess the consistency of these results in other cohorts of patients with or without impaired kidney function and to understand the mechanisms explaining these relationships.

Acknowledgments

The authors would like to thank Gigi Lipori (University of Florida Health and Shands Hospital, Gainesville, FL) and Carl Henriksen (College of Pharmacy, University of Florida, Gainesville, FL) for providing and cleaning electronic health data from UF Health Shands and UF Health Jacksonville.

Funding: None.

Competing interests: None declared.

Ethical approval: The study was approved by the institutional review and privacy boards of the University of Florida (Gainesville, FL).

Appendix. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.ijantimicag.2017.02.023.

References

- [1] Moellering RC Jr. Vancomycin: a 50-year reassessment. *Clin Infect Dis* 2006;42(Suppl. 1):S3–4.
- [2] Rybak M, Lomaestro B, Rotschafer JC, Moellering R Jr, Craig W, Billeter M, et al. Therapeutic monitoring of vancomycin in adult patients: a consensus review of the American Society of Health-System Pharmacists, the Infectious Diseases Society of America, and the Society of Infectious Diseases Pharmacists. *Am J Health Syst Pharm* 2009;66:82–98.
- [3] Bamgbola O. Review of vancomycin-induced renal toxicity: an update. *Ther Adv Endocrinol Metab* 2016;7:136–47.
- [4] Lau WK, Mercer D, Itani KM, Nicolau DP, Kuti JL, Mansfield D, et al. Randomized, open-label, comparative study of piperacillin–tazobactam administered by continuous infusion versus intermittent infusion for treatment of hospitalized patients with complicated intra-abdominal infection. *Antimicrob Agents Chemother* 2006;50:3556–61.
- [5] Burgess LD, Drew RH. Comparison of the incidence of vancomycin-induced nephrotoxicity in hospitalized patients with and without concomitant piperacillin–tazobactam. *Pharmacotherapy* 2014;34:670–6.
- [6] Gomes DM, Smotherman C, Birch A, Dupree L, Della Vecchia BJ, Kraemer DF, et al. Comparison of acute kidney injury during treatment with vancomycin in combination with piperacillin–tazobactam or cefepime. *Pharmacotherapy* 2014;34:662–9.
- [7] Peyko V, Smalley S, Cohen H. Prospective comparison of acute kidney injury during treatment with the combination of piperacillin–tazobactam and vancomycin versus the combination of cefepime or meropenem and vancomycin. *J Pharm Pract* 2017;30:209–13.
- [8] Karino S, Kaye KS, Navalkele B, Nishan B, Salim M, Solanki S, et al. Epidemiology of acute kidney injury among patients receiving concomitant vancomycin and piperacillin–tazobactam: opportunities for antimicrobial stewardship. *Antimicrob Agents Chemother* 2016;60:3743–50.
- [9] Moenster RP, Linneman TW, Finnegan PM, Hand S, Thomas Z, McDonald JR. Acute renal failure associated with vancomycin and β -lactams for the treatment of osteomyelitis in diabetics: piperacillin–tazobactam as compared with cefepime. *Clin Microbiol Infect* 2014;20:O384–9.
- [10] Hammond DA, Smith MN, Painter JT, Meena NK, Lusardi K. Comparative incidence of acute kidney injury in critically ill patients receiving vancomycin with concomitant piperacillin–tazobactam or cefepime: a retrospective cohort study. *Pharmacotherapy* 2016;36:463–71.
- [11] Davies SW, Efrid JT, Guidry CA, Dietrich ZC, Willis RN, Shah PM, et al. Top guns: the 'Maverick' and 'Goose' of empiric therapy. *Surg Infect (Larchmt)* 2016;17:38–47.
- [12] Kim T, Kandiah S, Patel M, Rab S, Wong J, Xue W, et al. Risk factors for kidney injury during vancomycin and piperacillin/tazobactam administration, including increased odds of injury with combination therapy. *BMC Res Notes* 2015;8:579.
- [13] Chawla LS, Kimmel PL. Acute kidney injury and chronic kidney disease: an integrated clinical syndrome. *Kidney Int* 2012;82:516–24.
- [14] McDonald JS, McDonald RJ, Lieske JC, Carter RE, Katzberg RW, Williamson EE, et al. Risk of acute kidney injury, dialysis, and mortality in patients with chronic kidney disease after intravenous contrast material exposure. *Mayo Clin Proc* 2015;90:1046–53.
- [15] Leonberg-Yoo AK, Tighiouart H, Levey AS, Beck GJ, Sarnak MJ. Urine potassium excretion, kidney failure, and mortality in CKD. *Am J Kidney Dis* 2017;69:341–9.
- [16] Hsu CY, Ordonez JD, Chertow GM, Fan D, McCulloch CE, Go AS. The risk of acute renal failure in patients with chronic kidney disease. *Kidney Int* 2008;74:101–7.
- [17] Waikar SS, Liu KD, Chertow GM. Diagnosis, epidemiology and outcomes of acute kidney injury. *Clin J Am Soc Nephrol* 2008;3:844–61.
- [18] Panwar B, Johnson VA, Patel M, Balkovetz DF. Risk of vancomycin-induced nephrotoxicity in the population with chronic kidney disease. *Am J Med Sci* 2013;345:396–9.
- [19] Andrassy KM. Comments on 'KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease'. *Kidney Int* 2013;84:622–3.
- [20] Khwaja A. KDIGO clinical practice guidelines for acute kidney injury. *Nephron Clin Pract* 2012;120:179–84.
- [21] Mamdani M, Sykora K, Li P, Normand SL, Streiner DL, Austin PC, et al. Reader's guide to critical appraisal of cohort studies: 2. Assessing potential for confounding. *BMJ* 2005;330:960–2.
- [22] Austin PC, Stuart EA. Moving towards best practice when using inverse probability of treatment weighting (IPTW) using the propensity score to estimate causal treatment effects in observational studies. *Stat Med* 2015;34:3661–79.
- [23] Brookhart MA, Wyss R, Layton JB, Sturmer T. Propensity score methods for confounding control in nonexperimental research. *Circ Cardiovasc Qual Outcomes* 2013;6:604–11.
- [24] Kellum JA, Murugan R. Effect of non-severe acute kidney injury on clinical outcomes in critically ill patients. *Crit Care* 2016;20:159.
- [25] Hoste EA, Bagshaw SM, Bellomo R, Cely CM, Colman R, Cruz DN, et al. Epidemiology of acute kidney injury in critically ill patients: the multinational AKI-EPI study. *Intensive Care Med* 2015;41:1411–23.
- [26] Forni LG, Dawes T, Sinclair H, Cheek E, Bewick V, Dennis M, et al. Identifying the patient at risk of acute kidney injury: a predictive scoring system for the development of acute kidney injury in acute medical patients. *Nephron Clin Pract* 2013;123:143–50.
- [27] Thakur CV, Christianson A, Almenoff P, Freyberg R, Render ML. Degree of acute kidney injury before dialysis initiation and hospital mortality in critically ill patients. *Int J Nephrol* 2013;2013:827459.
- [28] Sileanu FE, Murugan R, Lucko N, Clermont G, Kane-Gill SL, Handler SM, et al. AKI in low-risk versus high-risk patients in intensive care. *Clin J Am Soc Nephrol* 2015;10:187–96.
- [29] Lin J, Fernandez H, Shashaty MG, Negoianu D, Testani JM, Berns JS, et al. False-positive rate of AKI using consensus creatinine-based criteria. *Clin J Am Soc Nephrol* 2015;10:1723–31.
- [30] Pan HC, Wu PC, Wu VC, Yang YF, Huang TM, Shiao CC, et al. A nationwide survey of clinical characteristics, management, and outcomes of acute kidney injury (AKI)—patients with and without preexisting chronic kidney disease have different prognoses. *Medicine (Baltimore)* 2016;95:e4987.
- [31] Perazella MA, Luciano RL. Review of select causes of drug-induced AKI. *Expert Rev Clin Pharmacol* 2015;8:367–71.
- [32] Coca SG, Garg AX, Swaminathan M, Garwood S, Hong K, Thiessen-Philbrook H, et al. Preoperative angiotensin-converting enzyme inhibitors and angiotensin receptor blocker use and acute kidney injury in patients undergoing cardiac surgery. *Nephrol Dial Transplant* 2013;28:2787–99.
- [33] Solomon RJ, Natarajan MK, Doucet S, Sharma SK, Staniloae CS, Katholi RE, et al. Cardiac Angiography in Renally Impaired Patients (CARE) study: a randomized double-blind trial of contrast-induced nephropathy in patients with chronic kidney disease. *Circulation* 2007;115:3189–96.
- [34] Laskey W, Aspelin P, Davidson C, Rudnick M, Aubry P, Kumar S, et al. Nephrotoxicity of iodixanol versus iopamidol in patients with chronic kidney disease and diabetes mellitus undergoing coronary angiographic procedures. *Am Heart J* 2009;158:822–8, e3.
- [35] Chertow GM, Burdick E, Honour M, Bonventre JV, Bates DW. Acute kidney injury, mortality, length of stay, and costs in hospitalized patients. *J Am Soc Nephrol* 2005;16:3365–70.
- [36] Robins JM, Hernan MA, Brumback B. Marginal structural models and causal inference in epidemiology. *Epidemiology* 2000;11:550–60.